STATISTICS — MCQ ASSESSMENT

PART A — BASIC STATISTICS

1. What is the primary purpose of statistics?

A. To store data
B. To collect, analyze, interpret, and communicate data
C. To increase the size of a dataset
D. To convert all data into numbers

2. Which of the following is an example of quantitative data?

A. Eye color
B. Blood group
C. Height
D. Country

3. Which of the following is an example of categorical data?

A. Age
B. Weight
C. Temperature
D. Blood group

4. A population refers to:

A. A small subset of observations
B. The entire group being studied
C. Only the observations used for training
D. Only numerical observations

5. A sample is:

A. The entire population
B. A subset of the population
C. Always larger than the population
D. A single variable

6. Why is sampling commonly used?

A. Studying the entire population may be expensive or impractical
B. Samples always provide perfect results
C. Populations cannot be measured
D. Sampling eliminates all errors

PART B — MEASURES OF CENTRAL TENDENCY

7. Which measure represents the arithmetic average of a dataset?

A. Median
B. Mode
C. Mean
D. Range

8. What is the mean of:

10, 20, 30, 40, 50

A. 20
B. 25
C. 30
D. 35

9. The median is:

A. The largest value
B. The smallest value
C. The middle value after arranging the data
D. The most frequent value

10. What is the median of:

3, 7, 8, 10, 15

A. 7
B. 8
C. 10
D. 15

11. What is the median of:

2, 4, 6, 8

A. 4
B. 5
C. 6
D. 7

12. The mode represents:

A. The average value
B. The middle value
C. The most frequently occurring value
D. The largest value

13. Which dataset has a clear mode?

A. 1, 2, 3, 4, 5
B. 2, 2, 3, 4, 5
C. 1, 2, 3, 4, 5, 6
D. 10, 20, 30, 40

PART C — MEAN, MEDIAN AND OUTLIERS

14. Which measure is generally more sensitive to extreme values?

A. Mean
B. Median
C. Mode
D. Frequency

15. Consider:

10, 12, 13, 14, 100

Which measure would be most affected by the value 100?

A. Mean
B. Median
C. Mode
D. Minimum

16. If a very large outlier is added to a dataset, the mean will generally:

A. Decrease
B. Increase
C. Remain exactly the same
D. Become zero

17. For a highly skewed dataset containing extreme values, which measure may better represent the typical observation?

A. Mean
B. Median
C. Variance
D. Range

18. A dataset has mean = 50 and median = 50. What can we conclude?

A. The distribution must be perfectly normal
B. The distribution is necessarily symmetric
C. Mean and median happen to be equal
D. There are no outliers

PART D — MEASURES OF DISPERSION

19. What does dispersion describe?

A. The center of the data
B. How spread out the data are
C. The number of observations
D. The data type

20. Which of the following is a measure of dispersion?

A. Mean
B. Median
C. Standard deviation
D. Mode

21. The range of a dataset is:

A. Mean − Median
B. Maximum − Minimum
C. Maximum + Minimum
D. Median − Minimum

22. What is the range of:

5, 8, 10, 15, 20

A. 10
B. 15

C. 20
D. 25

23. If all values in a dataset are identical, the standard deviation is:

A. 0
B. 1
C. Equal to the mean
D. Infinite

24. A larger standard deviation generally indicates:

A. Less spread
B. Greater spread
C. Smaller mean
D. More observations

25. Two datasets have the same mean. Dataset A has standard deviation 2, while Dataset B has standard deviation 20. Which statement is most appropriate?

A. Dataset A has greater variability
B. Dataset B has greater variability
C. Both have identical variability
D. Standard deviation cannot compare datasets

PART E — VARIANCE

26. Variance is based on:

A. Absolute values from zero
B. Squared deviations from the mean
C. The maximum value only
D. The median

27. Why are deviations squared when calculating variance?

A. To make all deviations positive and emphasize larger deviations
B. To make the mean zero
C. To remove the observations
D. To reduce the dataset size

28. If the variance of a dataset is 25, its standard deviation is:

A. 5
B. 10
C. 25
D. 625

29. If the standard deviation is 4, the variance is:

A. 2
B. 4
C. 8
D. 16

30. Which statement is correct?

A. Variance and standard deviation always have the same units
B. Standard deviation is the square root of variance
C. Variance is the square root of standard deviation
D. Variance is always smaller than standard deviation

PART F — DISTRIBUTIONS

31. A distribution describes:

A. How values are arranged or occur in a dataset
B. Only the largest value
C. Only the average
D. The number of columns

32. A symmetric distribution has:

A. No mean
B. Similar shape on both sides of its center
C. Only positive values
D. No variation

33. In a perfectly symmetric distribution, the mean and median are generally:

A. Very different
B. Approximately equal
C. Always zero
D. Undefined

34. A distribution with a long tail toward larger values is described as:

A. Left-skewed
B. Right-skewed
C. Uniform
D. Symmetric

35. A distribution with a long tail toward smaller values is:

A. Right-skewed
B. Left-skewed
C. Uniform
D. Normal

36. In a positively skewed distribution, which relationship is generally expected?

A. Mean > Median
B. Mean < Median
C. Mean = 0
D. Median = 0

PART G — NORMAL DISTRIBUTION

37. A normal distribution is generally:

A. Symmetric and bell-shaped
B. Always skewed

C. Uniform
D. Discrete only

38. In an ideal normal distribution, the mean, median, and mode are:

A. Different
B. Approximately equal
C. Always zero
D. Undefined

39. The standard deviation of a distribution determines:

A. Its center only
B. Its spread
C. Its number of observations
D. Its maximum value

40. Increasing the standard deviation of a normal distribution generally makes the distribution:

A. Narrower
B. Wider
C. Shift to the right
D. Shift to the left

41. Decreasing the standard deviation while keeping the mean unchanged makes a normal distribution:

A. Wider
B. Narrower
C. More skewed
D. Shifted right

PART H — STANDARDIZATION AND Z-SCORE

42. A z-score indicates:

A. The original value of an observation
B. How far an observation is from the mean in standard deviation units
C. The number of observations
D. The variance of the dataset

43. If an observation has a z-score of 0, it is:

A. One standard deviation above the mean
B. At the mean
C. One standard deviation below the mean
D. At the maximum

44. A z-score of +2 means that the observation is:

A. 2 units above the mean
B. 2 standard deviations above the mean
C. 2 standard deviations below the mean
D. 2 units below the mean

45. A z-score of -3 means that the observation is:

A. 3 units below the mean
B. 3 standard deviations below the mean
C. 3 standard deviations above the mean
D. Equal to the mean

PART I — CORRELATION

46. Correlation measures:

A. The average of a variable
B. The strength and direction of association between two variables
C. The number of observations
D. The variance of one variable only

47. A correlation coefficient close to +1 indicates:

A. Strong positive linear association
B. Strong negative linear association
C. No linear association
D. Causation

48. A correlation coefficient close to -1 indicates:

A. Strong positive association
B. Strong negative linear association
C. No association
D. Identical variables

49. A correlation coefficient close to 0 generally indicates:

A. Strong positive linear relationship
B. Strong negative linear relationship
C. Little or no linear relationship
D. Perfect causation

50. If height and weight have a positive correlation, it means:

A. Increasing height necessarily causes weight to increase
B. Taller individuals tend to have higher weight
C. Every tall person is heavier
D. Height and weight are identical variables

51. Which statement about correlation is correct?

A. Correlation always implies causation
B. Correlation does not necessarily imply causation
C. Correlation can only be positive
D. Correlation can only be used for categorical variables

PART J — QUARTILES AND PERCENTILES

52. The median divides an ordered dataset into:

A. Two equal parts
B. Three equal parts
C. Four equal parts
D. Ten equal parts

53. The first quartile (Q1) represents approximately:

A. 10th percentile
B. 25th percentile
C. 50th percentile
D. 75th percentile

54. The third quartile (Q3) represents approximately:

A. 25th percentile
B. 50th percentile
C. 75th percentile
D. 90th percentile

55. The interquartile range (IQR) is:

A. Q1 + Q3
B. Q3 − Q1
C. Q3 × Q1
D. Q1 / Q3

56. Why is IQR useful?

A. It measures the spread of the middle 50% of the data
B. It measures the total number of observations
C. It always equals the standard deviation
D. It identifies the mean

PART K — SAMPLING

57. Random sampling means:

A. Selecting only the largest observations
B. Giving members of the population a chance of being selected
C. Selecting observations manually
D. Selecting only convenient observations

58. A sample that does not properly represent the population may lead to:

A. Sampling bias
B. Zero variance
C. Perfect prediction
D. Normal distribution

59. Increasing the sample size generally:

A. Makes estimates less reliable
B. Can make estimates more stable
C. Eliminates all sources of bias
D. Guarantees a normal population

60. Which of the following is most likely to produce a biased sample?

A. Randomly selecting people from the entire population
B. Selecting participants only from one convenient location

C. Using a properly designed random sampling method
D. Repeating random sampling several times

PART L — INTERPRETATION

61. A dataset has:

Mean = 50
Median = 50
Standard deviation = 2

Which statement is most reasonable?

A. The values are generally close to 50
B. The values are extremely spread out
C. The mean must be an outlier
D. There is no variation

62. A dataset has:

Mean = 50
Standard deviation = 30

What does this suggest?

A. The data have relatively high variability around the mean
B. All values are close to 50
C. All observations equal 50
D. The dataset has no variance

63. Two datasets have identical means and different standard deviations. What does this tell us?

A. They have exactly the same distribution
B. Their centers are the same, but their variability differs
C. Their medians must be different
D. Their sample sizes must be different

64. A dataset has a strong positive correlation between two variables. Which conclusion is safest?

A. One variable definitely causes the other
B. The variables have a strong positive linear association
C. The variables must have identical values
D. There is no relationship

65. Which statistic is generally most appropriate for describing the center of highly skewed data with extreme outliers?

A. Mean
B. Median
C. Variance
D. Standard deviation